

CS348 Computer Networks Quiz-1, Fri 29 Aug 2025, 08.35am-09.20am, Max. Marks: 10

Name: _____ Roll number: _____ House: _____

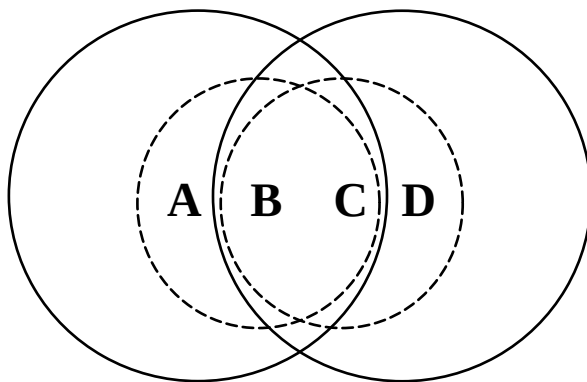
General instructions

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
- You are allowed to refer to your own hand-written notes only.
- All answers have to be (briefly) explained. State any necessary assumptions.

Prof Mitra Nischay (Mitran) wants to design a new wireless LAN technology, called MiNi (pronounced My Ny), as a competitor to WiFi. Your job is to help him answer the following questions.

1. **[1 mark]** Give one reason why MiNi should use FEC (Forward Error Correction) and not ARQ (Automatic Repeat reQuest).
2. **[1 mark]** Give one reason why MiNi should use ARQ and not FEC.
3. **[1 mark]** Prof Mitran has designed a link layer reliable transfer protocol with SWS=2 and RWS=3. Explain briefly as to why RWS=3 is a waste of buffer space. Draw a diagram as necessary.
4. **[1 mark]** Prof Mitran designed a sequence number space of size 3 (i.e. 0, 1, 2 repeating). Explain through a suitable timing diagram as to why this will lead to incorrect behaviour. Timing diagram is mandatory.

5. **[2 marks]** Prof Mitran has a brilliant student Sahasrabuddhi who has designed a PHY layer with capacity of 16 Mbps. But instead of using it in a LAN, Prof Mitran struck a deal with ISRO to use it to communicate with a LEO satellite. Suppose the above reliable transfer protocol is used with a frame size of 200 bytes, and the satellite is at a distance of 150km from ISRO's base on earth. What is the best case efficiency of this reliable transfer protocol? Ignore header overheads and ACK size. Show your calculation briefly.
6. **[1 mark]** Sahasrabuddhi's friend Ekabuddhi test deployed 4 nodes of MiNi in his office building. The diagram below shows the transmission ranges of the 4 nodes. The solid circles represent the transmission range of nodes A and D, and the dashed circles represent the transmission range of B and C.



The absent minded Prof Mitran forgot to design a MAC protocol for MiNi. Suppose A is sending data to B, while C is sending data to D. Assume both are sending packets continuously at a rate equal to the link capacity, and without any ACK mechanism. What is the received throughput at B as a percentage of A's sending rate? Explain briefly.

7. **[1 mark]** In the above scenario, what is the received throughput at D as a percentage of C's sending rate? Explain briefly.
8. **[2 marks]** Prof Mitran had another clever and loyal student Kotibuddhi, who said that Prof Mitran's absent mindedness (no MAC in MiNi) is a blessing in disguise. She described a scenario of two nodes transmitting in the above network (again without any ACK mechanism), while achieving twice the overall throughput compared to the case when CSMA/CA is used. Can you match Kotibuddhi's brains? Explain briefly.